

Algebra Essentials

ANSWER KEY



Evaluating expressions

- 1 If $x = 4$, evaluate $3x + 7$.
 $3(4) + 7 = 12 + 7 = 19$.
 - 2 If $a = 5$ and $b = 2$, evaluate $a^2 - 3b$.
 $5^2 - 3(2) = 25 - 6 = 19$.
 - 3 If $x = -3$, evaluate $2x^2 + 5x$.
 $2(-3)^2 + 5(-3) = 2(9) - 15 = 18 - 15 = 3$.
 - 4 If $p = 6$ and $q = -2$, evaluate $\frac{p-q}{q+6}$.
 $\frac{6 - (-2)}{-2 + 6} = \frac{8}{4} = 2$.
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Solving linear equations

- 5 Solve $2x + 5 = 17$.
 $2x = 12$, so $x = 6$.
 - 6 Solve $5x - 8 = 27$.
 $5x = 35$, so $x = 7$.
 - 7 Solve $3(x - 4) = 21$.
 $x - 4 = 7$, so $x = 11$.
 - 8 Solve $\frac{x}{4} + 3 = 10$.
 $\frac{x}{4} = 7$, so $x = 28$.
 - 9 Solve $7x + 3 = 4x + 18$.
 $7x - 4x = 18 - 3$, so $3x = 15$, and $x = 5$.
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Expanding and simplifying

- 10 Expand $(x + 3)(x + 5)$.
 $x^2 + (3 + 5)x + 3 \times 5 = x^2 + 8x + 15$.
- 11 Expand $(x - 4)(x + 6)$.
 $x^2 + (-4 + 6)x + (-4)(6) = x^2 + 2x - 24$.

- 12 Expand $(2x + 1)(x - 3)$.
 $2x^2 - 6x + x - 3 = 2x^2 - 5x - 3$.
- 13 Simplify $3(2x - 5) - 2(x - 4)$.
 $6x - 15 - 2x + 8 = 4x - 7$.
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Factoring

- 14 Factor $x^2 + 9x + 20$.
Find two numbers with sum 9 and product 20: 4 and 5. $(x + 4)(x + 5)$.
- 15 Factor $x^2 - 2x - 15$.
Find two numbers with sum -2 and product -15 : -5 and 3 . $(x - 5)(x + 3)$.
- 16 Factor $x^2 - 49$.
Difference of squares: $(x - 7)(x + 7)$.
- 17 Factor $6x + 15$.
Common factor 3: $3(2x + 5)$.
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Solving quadratic equations by factoring

- 18 Solve $x^2 + 5x + 6 = 0$.
Factor: $(x + 2)(x + 3) = 0$, so $x = -2$ or $x = -3$.
- 19 Solve $x^2 - 7x + 12 = 0$.
Factor: $(x - 3)(x - 4) = 0$, so $x = 3$ or $x = 4$.
- 20 Solve $x^2 - 16 = 0$.
Difference of squares: $(x - 4)(x + 4) = 0$, so $x = 4$ or $x = -4$.
- 21 Solve $x^2 + 2x - 15 = 0$.
Factor: $(x + 5)(x - 3) = 0$, so $x = -5$ or $x = 3$.
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Systems of two equations

- 22 Solve the system: $x + y = 10$, $x - y = 4$.
Adding the equations: $2x = 14$, so $x = 7$. Then $y = 10 - 7 = 3$.
- 23 Solve the system: $2x + y = 11$, $x - y = 1$.
Adding the equations: $3x = 12$, so $x = 4$. Then $y = x - 1 = 3$.

- 24** Solve the system: $x = 3y - 1$, $2x + y = 19$.
Substitute: $2(3y - 1) + y = 19 \Rightarrow 6y - 2 + y = 19 \Rightarrow 7y = 21 \Rightarrow y = 3$. Then $x = 3(3) - 1 = 8$.
- 25** Solve the system: $4x - y = 10$, $2x + y = 8$.
Adding the equations: $6x = 18$, so $x = 3$. Then $y = 8 - 2(3) = 2$.
- 26** Solve the system: $5x + 3y = 21$, $2x - 3y = 0$.
Adding the equations: $7x = 21$, so $x = 3$. Then $3y = 2x = 6$, so $y = 2$.
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Laws of exponents

- 27** Simplify $a^5 \times a^3$.
 $a^{5+3} = a^8$.
- 28** Simplify $\frac{x^9}{x^4}$.
 $x^{9-4} = x^5$.
- 29** Simplify $(y^3)^4$.
 $y^{3 \times 4} = y^{12}$.
- 30** Simplify $2x^3 \times 5x^2$.
 $2 \times 5 \times x^{3+2} = 10x^5$.
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Inequalities

- 31** Solve $3x - 4 > 11$.
 $3x > 15$, so $x > 5$.
- 32** Solve $-2x + 6 \leq 20$.
 $-2x \leq 14$. Dividing by a negative reverses the inequality: $x \geq -7$.
- 33** Solve $5x + 1 \geq 3x + 9$.
 $2x \geq 8$, so $x \geq 4$.
- 34** Solve $4(x - 2) < 2x + 6$.
 $4x - 8 < 2x + 6 \Rightarrow 2x < 14 \Rightarrow x < 7$.
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Substitution into formulas

- 35** The formula for simple interest is $I = Prt$. Find I when $P = 2,000$, $r = 0.04$, $t = 3$.
 $I = 2,000 \times 0.04 \times 3 = 240$.

- 36 The perimeter of a rectangle is $P = 2(l + w)$. Find P when $l = 15$ and $w = 9$.
 $P = 2(15 + 9) = 2(24) = 48$.
- 37 If $v = u + at$, find v when $u = 5$, $a = 3$, $t = 4$.
 $v = 5 + 3(4) = 5 + 12 = 17$.
- 38 The area of a triangle is $A = \frac{1}{2}bh$. Find A when $b = 14$ and $h = 9$.
 $A = \frac{1}{2} \times 14 \times 9 = 63$.
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Translating words into equations

- 39 Twice a number, plus 7, equals 25. Find the number.
 $2x + 7 = 25 \Rightarrow 2x = 18 \Rightarrow x = 9$.
- 40 A number decreased by 8 is equal to three times the number, decreased by 20. Find the number.
 $x - 8 = 3x - 20 \Rightarrow 12 = 2x \Rightarrow x = 6$.
- 41 The sum of three consecutive integers is 84. Find the integers.
Let the integers be n , $n + 1$, $n + 2$: $3n + 3 = 84 \Rightarrow 3n = 81 \Rightarrow n = 27$. The integers are 27, 28, 29.
- 42 Five more than three times a number is the same as 10 less than five times the number. Find the number.
 $3x + 5 = 5x - 10 \Rightarrow 15 = 2x \Rightarrow x = 7.5$.
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Word problems

- 43 A rectangle has length $(x + 3)$ cm and width x cm. This question has two parts.
- a If the area is 40 cm², write an equation in x . Area = $x(x + 3) = 40$, i.e. $x^2 + 3x - 40 = 0$.
- b Solve for x . Factoring: $(x + 8)(x - 5) = 0$. Since length must be positive, $x = 5$.
- 44 A father is 3 times as old as his son. In 12 years, he will be twice as old as his son. Find their current ages.
Let the son's age be x ; the father's is $3x$. In 12 years:
 $3x + 12 = 2(x + 12) \Rightarrow 3x + 12 = 2x + 24 \Rightarrow x = 12$. Son is 12, father is 36.
- 45 Tickets cost 25 riyals for adults and 15 riyals for children. A group of 10 people paid 220 riyals in total. How many adults were in the group?
Let a be the number of adults, so $10 - a$ are children:
 $25a + 15(10 - a) = 220 \Rightarrow 25a + 150 - 15a = 220 \Rightarrow 10a = 70 \Rightarrow a = 7$ adults.
- 46 The sum of two numbers is 45, and their difference is 11. Find the two numbers.
Let the numbers be x and y with $x + y = 45$ and $x - y = 11$. Adding: $2x = 56 \Rightarrow x = 28$, then $y = 17$.